



Unit 3 · Outcome 1 · Maintaining Biodiversity

Nobody planted the bush. The planet did.

KK05

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



One idea, five forces

01 One idea, five forces Everything in this dot point hangs off a single sentence — learn it and you can rebuild the rest. ♦ The sentence to memorise Natural drivers change ecosystems over different time scales , and in doing so they create endemism , concentrate it into biodiversity hotspots , and drive both extinction and new diversification . The five drivers VCAA names are volcanic eruptions , fire , droughts and floods linked to El Niño , plate tectonic movement , and evolution / natural selection . Two of these are slow giants working over millions

"Different time scales" — the examiner's favourite phrase

02 "Different time scales" — the examiner's favourite phrase Place each driver on the line. Short-term drivers reshuffle who's present; long-term drivers change what exists at all. El Niño / La Niña 2–7 yrs Fire regime years–decades Eruption + succession decades–centuries Evolution 10^3 – 10^7 yrs Plate tectonics 10^7 + yrs ◀ short-term · ecological geological · long-term ▶ ♦ Why the distinction earns marks A short-term driver like an El Niño drought changes which species are present right now (population crashes, range shifts). A long-term driver like continue

The five drivers, up close

03 The five drivers, up close Each card pairs the driver with a real Australian example you can quote in an answer. Volcanic eruptions event → succession over decades–centuries Wipe an area to bare rock, then trigger primary succession — new, fertile, isolated habitat that can later host unique species. Fire years–decades (fire regimes) s vegetation. The right fire regime maintains diversity and drives speciation; the wrong one erases it. El Niño droughts & floods 2–7 year cycles Swings rainfall. Droughts bleach reefs and shrink populations; La Niñ

Three words the drivers produce

04 Three words the drivers produce Endemism, hotspots and extinction aren't separate topics — they're the outputs of the drivers above. Endemism made mostly by isolation A species found in one region and nowhere else . Driven by isolation from plate tectonics, islands and mountains. Australian examples → Biodiversity hotspot high endemism + high threat A region with exceptional endemism that is also highly threatened . SW Australia → Extinction natural background + driver-spikes Loss of a species forever. Natural drivers cause it (volcanic winters

Deep-time explorer

05 Deep-time explorer Tap a driver on the time axis. The further right, the slower it works — and the deeper its effect on what exists. Click a driver to read its time scale & effect click a glowing marker Loading time axis... SELECT A DRIVER Tap a marker → time scale The five drivers sit on a logarithmic time axis from a few years to hundreds of millions of years. The markers nearest you act fastest. Acts over... — Pick a driver to see a quotable Australian example.

Disturbance & recovery

06 Disturbance & recovery Hit the ecosystem with a natural driver and watch diversity crash — then recover through succession. Fire and eruption behave differently. That difference is the point. Run a disturbance, watch the curve live diversity readout → Growing the ecosystem... Bushfire Eruption ↺ Regrow Live readout Species diversity 100% Biomass / cover 100% Mature ecosystem. Hit it with a disturbance to begin. ♦ Read the curve like an examiner A bushfire in a fire-adapted system causes a sharp dip then fast recovery — seed banks, lignotubers and

Sort the drivers

◇ ACTIVE RECALL

07 Sort the drivers Drag (or tap) each example into the time scale it works over. Short-term reshuffles who's present; long-term changes what exists. Short-term · ecological (years–centuries) changes which species are present now Long-term · geological (thousands–millions of yrs) changes which species exist at all s

Decode the command word

◆ COMMAND WORDS

08 Decode the command word Half of losing marks is answering the wrong verb. Tap a card to see what the examiner actually wants for this topic.

Exam traps to dodge

⚠ EXAM TRAPS

09 Exam traps to dodge The mistakes that quietly cost marks on this dot point.

Writing trainer · P·E·L

✍ P·E·L WRITING

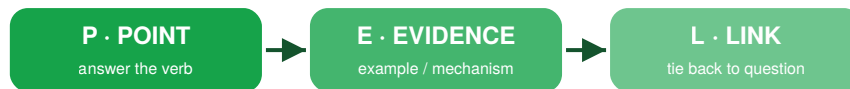
10 Writing trainer · P·E·L A reliable structure for "explain / analyse" answers: Point → Evidence → Link . Open each step to build a model paragraph. 4 marks Explain how plate tectonic movement has contributed to high levels of endemism in Australia. Refer to time scale.



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Good answer vs bad answer

11 Good answer vs bad answer Same question, two responses. The highlights show exactly where marks are won and lost.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

Practice questions

★ PRACTICE & SELF-MARK

12 Practice questions Write your answer first, then reveal the model, tick off the marking points, and self-mark. Your total tracks in the score dock.

60 -second recap

♦ RECAP / CHEAT SHEET

13 60-second recap If you remember five things Five natural drivers: volcanic eruptions, fire, El Niño droughts/floods, plate tectonics, and evolution. Different time scales: fire & ENSO act over years–decades (which species are present); plate tectonics & evolution act over millions of years (which species exist). Endemism is mainly built by isolation — and Australia's drift from Gondwana makes it a world leader in endemic species. A hotspot = high endemism + high threat (e.g. the South-West Australian Floristic Region). Drivers cause extinction and d